Chapter 7 • Coordinate Geometry

Solved Exercise 7.2

1. Find the slope and inclination of the line joining each pair of points.

(i) Through $(-2,4)$ and $(5,11)$: $m=\frac{11-4}{5+2}=\frac{7}{7}=1$, so $\alpha=\tan^{-1}1=45°$.

$m=1, \alpha=45°$

(ii) Through $(3,-2)$ and $(2,7)$: $m=\frac{7+2}{2-3}=\frac{9}{-1}=-9$. The line lies in Quadrant II direction, so $\alpha=180°-\tan^{-1}9\approx 96.340°=96°20'$ approximately.

$m=-9, \alpha\approx 96°20'$

(iii) Through $(4,6)$ and $(4,8)$: $x_2-x_1=0$, so the line is vertical, its slope is undefined and $\alpha=90°$.

slope undefined, $\alpha=90°$.

2. By means of slopes, show that each triple of points is collinear.

(i) $A(-1,-3), B(1,5), C(2,9)$: $m_{AB}=\frac{5+3}{1+1}=4$ and $m_{BC}=\frac{9-5}{2-1}=4$. Equal slopes through B prove collinearity.

(ii) $P(4,-5), Q(7,5), R(10,15)$: $m_{PQ}=\frac{5+5}{7-4}=\frac{10}{3}$ and $m_{QR}=\frac{15-5}{10-7}=\frac{10}{3}$

(iii) $L(-4,6), M(3,8), N(10,10)$: $m_{LM}=\frac{8-6}{3+4}=\frac{2}{7}$ and $m_{MN}=\frac{10-8}{10-3}=\frac{2}{7}$

(iv) $X(a,2b), Y(c,a+b), Z(2c-a,2a)$: $m_{XY}=\frac{a-b}{c-a}$ and $m_{YZ}=\frac{a-b}{c-a}$. If $c=a$, all three points instead lie on the vertical line $x=a$.

3. Find k for AB parallel or perpendicular to CD, where A(7,3), B(k,−6), C(−4,5), D(−6,4).

$m_{CD}=\frac{4-5}{-6+4}=\frac{1}{2}$ and $m_{AB}=\frac{-9}{k-7}$.

(i) Parallel: $\frac{-9}{k-7}=\frac{1}{2}$

$\Rightarrow k-7=-18$.

$k=-11$

(ii) Perpendicular: $m_{AB}=-2 \Rightarrow \frac{-9}{k-7}=-2 \Rightarrow k-7=\frac{9}{2}$

$k=\frac{23}{2}$

4. Show A(6,1), B(2,7), C(−6,−7) form a right triangle.

$m_{AB}=\frac{7-1}{2-6}=\frac{-3}{2}$ and $m_{AC}=\frac{-7-1}{-6-6}=\frac{2}{3}$ Their product is -1, so $AB \perp AC$ and the right angle is at A.

5. Determine whether each pair of lines is parallel, perpendicular or neither.

(a) Through $(1,-2),(2,4)$: $m_1=\frac{4+2}{2-1}=6$. Through $(4,1),(-8,2)$: $m_2=\frac{2-1}{-8-4}=\frac{-1}{12}$

Since $6\neq -\frac{1}{12}$ and $6(\frac{-1}{12})=\frac{-1}{2}\neq -1$, the lines in (a) are neither parallel nor perpendicular.

neither

(b) Through $(-3,4),(6,2)$: $m_1=\frac{2-4}{6+3}=\frac{-2}{9}$ Through $(4,5),(-2,-7)$: $m_2=\frac{-7-5}{-2-4}=2$.

Since the slopes are unequal and their product is $\frac{-4}{9}$, not -1, the lines in (b) are also neither.

6. Find the equation of each described line.

(a) Horizontal through $(7,-9)$: $y=-9$, or $y+9=0$.
(b) Vertical through $(-5,3)$: $x=-5$, or $x+5=0$.
(c) Through $A(-6,5)$ with slope 7: $y-5=7(x+6) \Rightarrow 7x-y+47=0$.
(d) Through $(8,-3)$ with slope 0: $y+3=0(x-8)$, so $y+3=0$.
(e) Through $(-8,5)$ with undefined slope: the line is vertical, so $x+8=0$.
(f) Through $(-5,-3)$ and $(9,-1)$: $m=\frac{-1+3}{9+5}=\frac{1}{7}$ Thus $y+3=\frac{x+5}{7} \Rightarrow x-7y-16=0$.
(g) y-intercept -7 and slope -5: $y=-5x-7 \Rightarrow 5x+y+7=0$.
(h) x-intercept -3 and y-intercept 4: $\frac{x}{-3}\frac{+y}{4}=1 \Rightarrow 4x-3y+12=0$.
(i) x-intercept -9 and slope -4: use $(-9,0)$, so $y=-4(x+9) \Rightarrow 4x+y+36=0$.

7. Find the perpendicular bisector of A(3,5), B(9,8).

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Midpoint=$(6,\frac{13}{2})$; segment slope=$\frac{8-5}{9-3}=\frac{1}{2}$, so perpendicular slope=-2.

$y-\frac{13}{2}=-2(x-6)$

$\Rightarrow 2x+y=\frac{37}{2}$

$4x+2y-37=0$.

8. Line through $(-4,-6)$, perpendicular to slope $\frac{-3}{2}$

Required slope=$\frac{2}{3}$: $y+6=(\frac{2}{3})(x+4)$.

$2x-3y-10=0$.

9. Line through $(11,-5)$, parallel to slope -24.

$y+5=-24(x-11)$.

$24x+y-259=0$.

10. Convert each equation into slope-intercept form $y=mx+c$, two-intercept form $x/a+y/b=1$ and normal form $x\cos\alpha+y\sin\alpha=p$.

(a) $2x-4y+11=0$: $4y=2x+11$ gives $y=\frac{x}{2}\frac{+11}{4}$ The intercepts are $\frac{-11}{2}$ and $\frac{11}{4}$, so $\frac{x}{\frac{-11}{2}}\frac{+y}{\frac{11}{4}}=1$.

Divide $2x-4y=-11$ by $-\sqrt{20}$ to make p positive: $(\frac{-1}{\sqrt{5}})x+\frac{2}{\sqrt{5}}y=\frac{11}{2\sqrt{5}}$.

(b) $4x+7y-2=0$: $y=\frac{-4x}{7}\frac{+2}{7}$ The intercepts are $\frac{1}{2}$ and $\frac{2}{7}$, so $\frac{x}{\frac{1}{2}}\frac{+y}{\frac{2}{7}}=1$. Divide $4x+7y=2$ by $\sqrt{65}$:

$(\frac{4}{\sqrt{65}})x+\frac{7}{\sqrt{65}}y=\frac{2}{\sqrt{65}}$

(c) $15y-8x+3=0$: $y=\frac{8x}{15}\frac{-1}{5}$ The intercepts are $\frac{3}{8}$ and $\frac{-1}{5}$, so $\frac{x}{\frac{3}{8}}\frac{+y}{\frac{-1}{5}}=1$. Divide $-8x+15y=-3$

by -17: $(\frac{8}{17})x-\frac{15}{17}y=\frac{3}{17}$

11. Check whether each pair of lines is parallel, perpendicular or neither.

(a) $2x+y-3=0$ has slope -2; $4x+2y+5=0$ also has slope -2. They are distinct parallel lines.

parallel

(b) $3y=2x+5$ has slope $\frac{2}{3}$; $3x+2y-8=0$ has slope $\frac{-3}{2}$ Their product is -1, so they are perpendicular.

(c) $4y+2x-1=0$ has slope $\frac{-1}{2}$; $x-2y-7=0$ has slope $\frac{1}{2}$ The slopes are unequal and their product is $\frac{-1}{4}$, so neither.

12. Line through $(-4,7)$, parallel to $2x-7y+4=0$.

Required slope=$\frac{2}{7}$: $y-7=(\frac{2}{7})(x+4)$.

$2x-7y+57=0$.

13. Line through $(5,-8)$, perpendicular to the join of $A(-15,-8)$, $B(10,7)$.

Join slope=$\frac{7+8}{10+15}=\frac{3}{5}$; required slope=$\frac{-5}{3}$

$y+8=-(\frac{5}{3})(x-5)$

$\Rightarrow 3y+24=-5x+25$.

$5x+3y-1=0$.

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